

Build a Smart Parking System for a Metro Station

This article will help you design a Web based application that automates a smart parking system for cars in a metro station using Node-RED.



A Web application is software that gets executed on a Web server. Every Web application is accessed by the end user through a Web browser. These Web applications are programmed using a client-server architecture, where the user (client) is provided services through a remotely located server that might be hosted by a third-party. A Web API (application programming interface) is available across the Web and can be accessed by the user using the HTTP protocol, as shown in Figure 1.

This article will demonstrate how to design a Web based application for an automated smart parking system for cars in a metro station. It is designed using open source Node-RED. This system creates an interactive and stylish user login form using a template node, where HTML and CSS are coded to get car owner details to automate the parking system. We can see the login and submission form flow diagrams in Figures 2 and 3.

The nodes used are as follows:

Http in node	
Function node	
Template node	
Http response node	
JSON node	
Debug node	
Comment node	

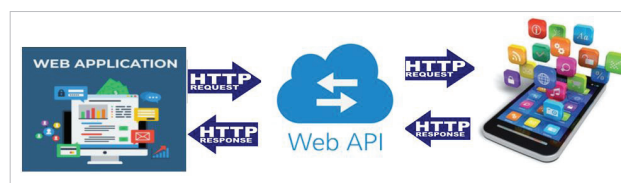


Figure 1: Web API

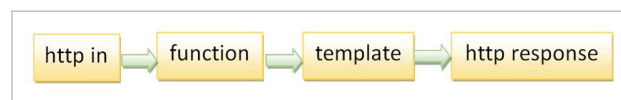


Figure 2: Login form flow diagram

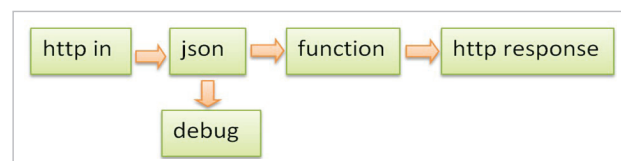


Figure 3: Submission form flow diagram